A CHARACTERISATION OF SEPARATED FORMAL ALGEBRAIC SPACES THROUGH FIBRE PRODUCTS OF AFFINES

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The characterisation of quasi-separated algebraic spaces through fibre products of affines and the surjectivity of the comparison map. Properties of Spaces, Lemma spaces-properties-lemma-characterize-quasi-separated, invoked at proof lines 3628-3631 and 3645-3649. Gives the easy direction and the affineness of the reduction of the fibre product.
- The reduction of a formal algebraic space and the base change identity for fibre products of reductions. Same chapter, Lemma lemma-reduction-formal-algebraic-space at statement lines 3612-3613 and Lemma lemma-reduction-fibre-products at proof lines 3639-3642. Identifies the reduction of the fibre product with the reduction of the fibre product of reductions.
- The fibre product of two affines over the space is a scheme, and the diagonal is a monomorphism locally of finite type. Same chapter, Lemma lemma-diagonal-formal-algebraic-space, invoked at proof lines 3635-3637 and 3662-3665. Makes the fibre product a scheme and gives the first factor of the comparison morphism.
- An affine algebraic space whose reduction is affine is affine, and an integral morphism is universally closed. Limits of Spaces, Proposition spaces-limits-proposition-affine at proof lines 3649-3650, and Morphisms, Lemma morphisms-lemma-integral-universally-closed at proof lines 3668-3670. Give affineness of the fibre product and integrality of the comparison morphism.
- An integral finite type epimorphism of rings is finite and hence surjective. Morphisms, Lemma morphisms-lemma-finite-integral and Algebra, Lemma algebra-lemma-finite-epimorphism-surjective, invoked at proof lines 3679-3683. Concludes the surjectivity of the comparison map.

Date: stacks-project:formal-spaces.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 04VT: sources/stacks-chow/tags.txt, source lines 6336–6336. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 0AIP: sources/stacks-formal-spaces/pinned-original-source.tex, source lines 2667–2667. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 0AIN: sources/stacks-formal-spaces/pinned-original-source.tex, source lines 2833–2833. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 01WJ: sources/stacks-chow/tags.txt, source lines 2484–2484. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 01WM: sources/stacks-chow/tags.txt, source lines 2487–2487. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 01KR: sources/stacks-chow/tags.txt, source lines 2060–2060. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original proposition 07VT: sources/stacks-chow/tags.txt, source lines 10018–10018. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 0AHR: sources/stacks-chow/tags.txt, source lines 13239–13239. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Lemma 0.1. *Let S be a scheme. Let X be a formal algebraic space over S. The following are equivalent*

1. *the reduction of X (Lemma 0AIN) is a separated algebraic space,*
2. *for $U \to X$, $V \to X$ with U, V affine the fibre product $U \times_X V$ is affine and*
$$\mathcal{O}(U) \otimes_{\mathbf{Z}} \mathcal{O}(V) \longrightarrow \mathcal{O}(U \times_X V)$$
is surjective.

Proof. If (2) holds, then X_{red} is a separated algebraic space by applying Properties of Spaces, Lemma 0AHR to morphisms $U \to X_{red}$ and $V \to X_{red}$ with U, V affine and using that $U \times_{X_{red}} V = U \times_X V$.

Assume (1). Let $U \to X$ and $V \to X$ be as in (2). Observe that $U \times_X V$ is a scheme by Lemma 0AIP. Let $U_{red}, V_{red}, X_{red}$ be the reduction of U, V, X. Then

$$U_{red} \times_{X_{red}} V_{red} = U_{red} \times_X V_{red} \to U \times_X V$$

is a thickening of schemes. It follows that $(U \times_X V)_{red} = (U_{red} \times_{X_{red}} V_{red})_{red}$. In particular, we see that $(U \times_X V)_{red}$ is an affine scheme and that

$$\mathcal{O}(U) \otimes_{\mathbf{Z}} \mathcal{O}(V) \longrightarrow \mathcal{O}((U \times_X V)_{red})$$

is surjective, see Properties of Spaces, Lemma 0AHR. Then $U \times_X V$ is affine by Limits of Spaces, Proposition 07VT. On the other hand, the morphism $U \times_X V \to U \times V$ of affine schemes is the composition

$$U \times_X V = X \times_{(X \times_S X)} (U \times_S V) \to U \times_S V \to U \times V$$

The first morphism is a monomorphism and locally of finite type (Lemma 0AIP). The second morphism is an immersion (Schemes, Lemma 01KR). Hence the composition is a monomorphism which is locally of finite type. On the other hand, the composition is integral as the map on underlying reduced affine schemes is a closed immersion by the above and hence universally closed (use Morphisms, Lemma 01WM). Thus the ring map

$$\mathcal{O}(U) \otimes_{\mathbf{Z}} \mathcal{O}(V) \longrightarrow \mathcal{O}(U \times_X V)$$

is an epimorphism which is integral of finite type hence finite hence surjective (use Morphisms, Lemma 01WJ and Algebra, Lemma 04VT). □